IPv4 Addressing

Understanding How Devices Talk on the Internet

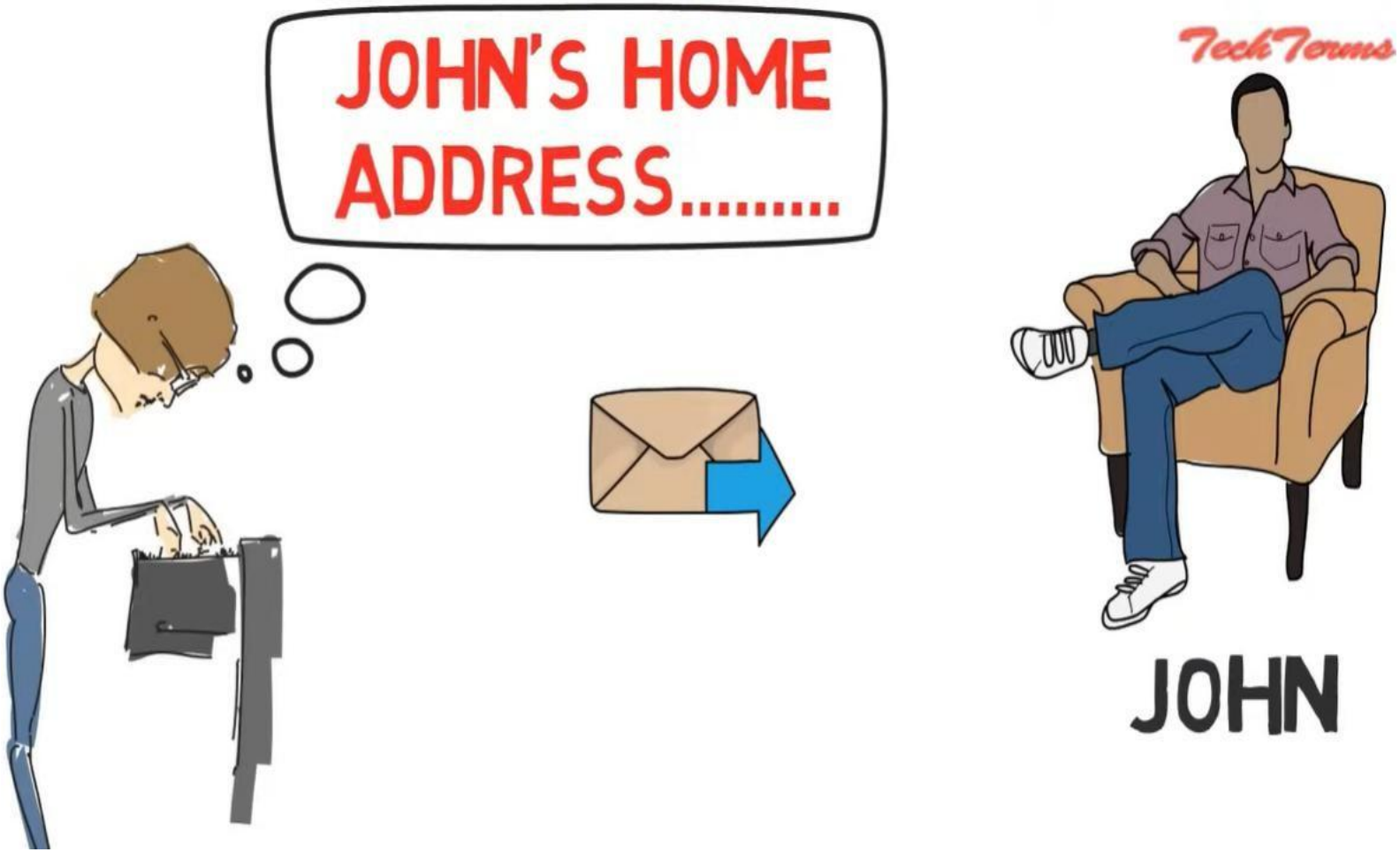
Opening Story

STORY

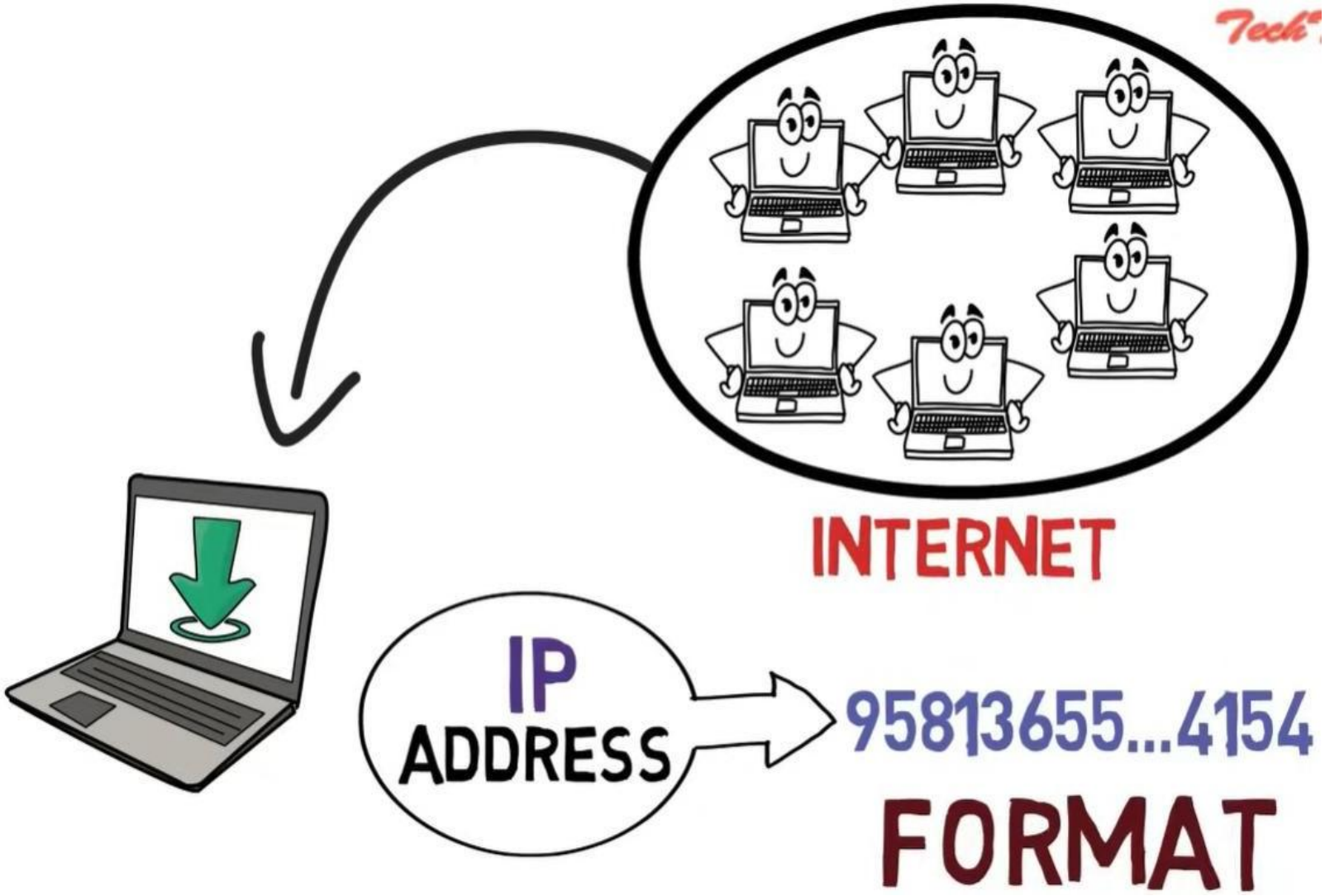
"A City Without House Numbers"

Imagine a city where no house has a number. A delivery person arrives with a package but has no idea where to deliver it.?

Opening Story



Opening Story

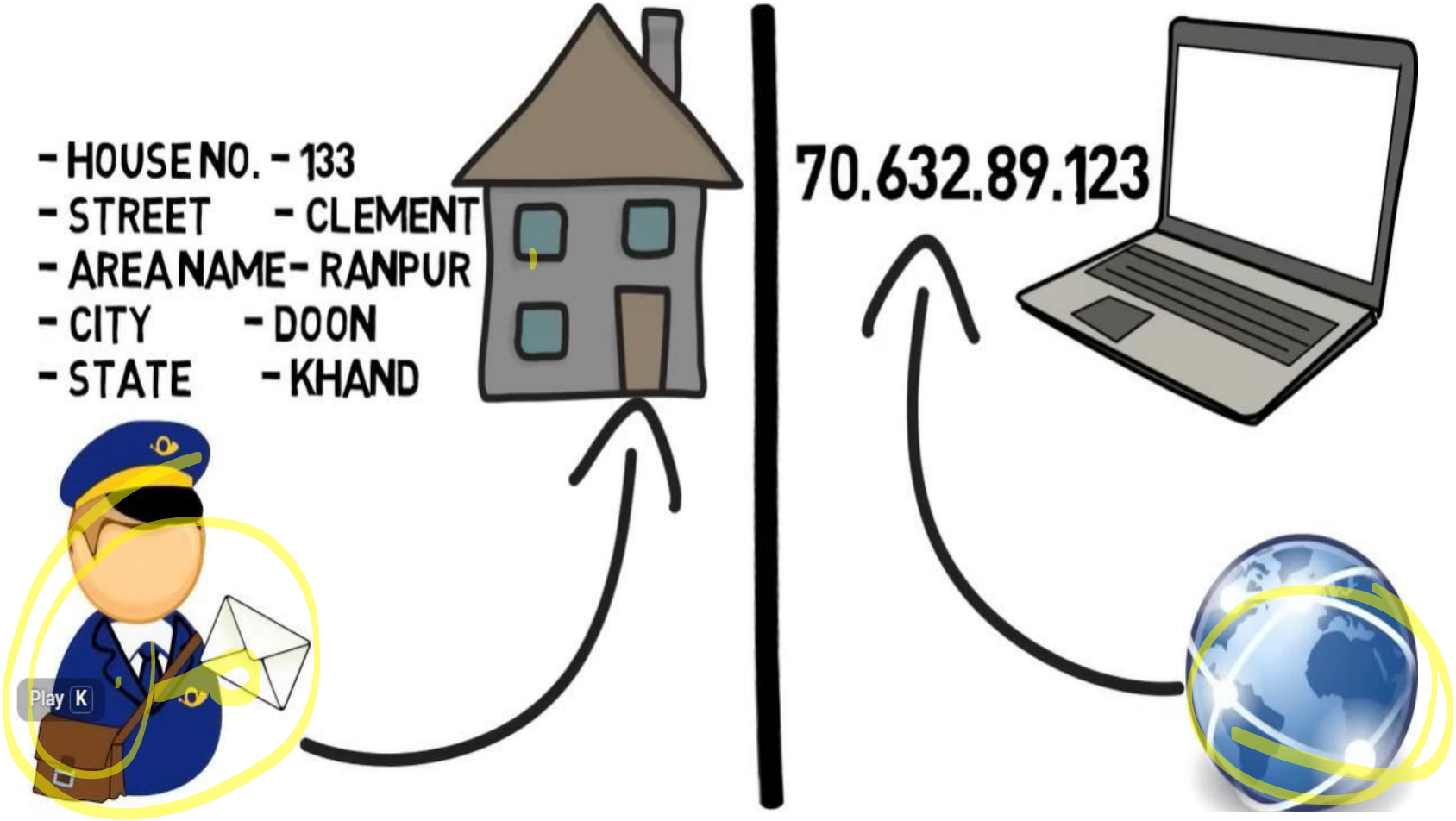


Opening Story

- The internet once faced a similar problem. Millions of devices wanted to communicate, but without a proper addressing system, data packets had nowhere to go.

- **That's where IPv4 comes in -** it gives every device a unique address, just like a house number in a city.

Opening Story

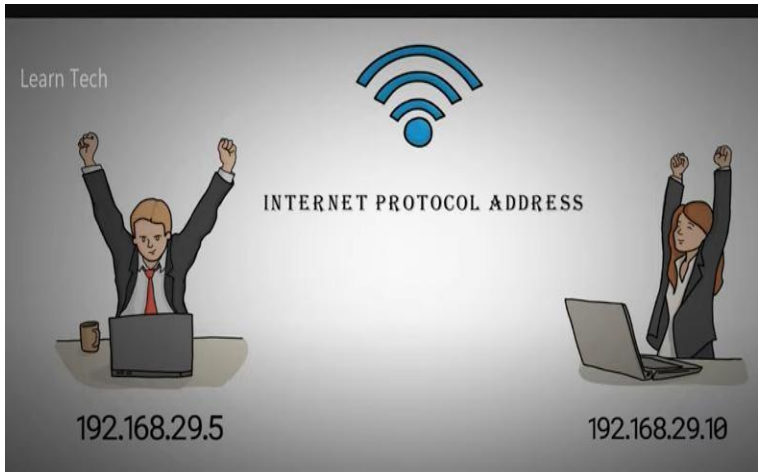


What is IPv4?

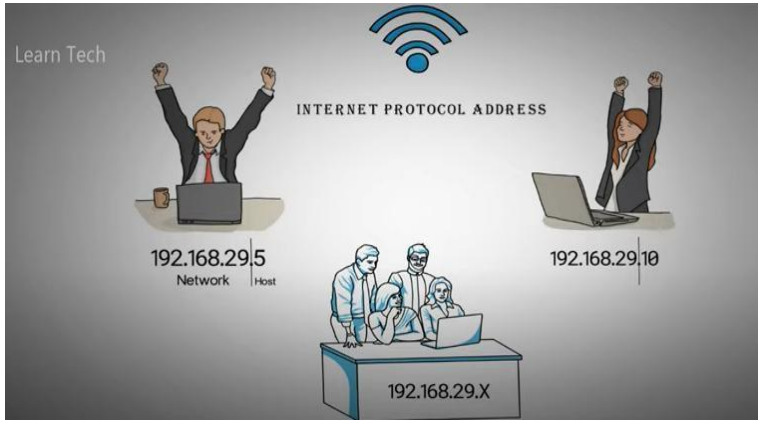
- IPv4 = Internet Protocol Version 4
- Logical addressing system used to identify devices on a network
- 32-bit address
- Written in **dotted decimal format**

Example: 192.168.1.1

IP address


Learn Tech
INTERNET PROTOCOL ADDRESS
192.168.29.5
192.168.29.10

192.168.29.5
Network
Host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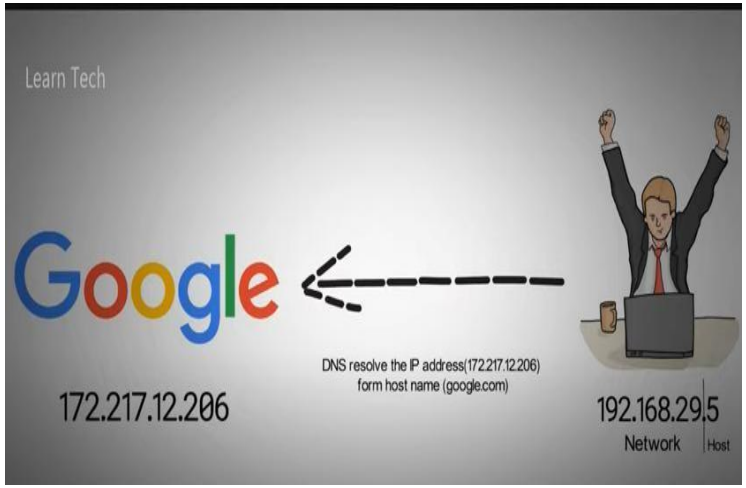
INTERNET PROTOCOL ADDRESS
192.168.29.5
Network
Host
192.168.29.10
192.168.29.X


Learn Tech
Google
DNS resolve the IP address(172.217.12.206)
form host name (google.com)
172.217.12.206
192.168.29.5
Network
Host

IP address

- IPV4 Address
- IPV6 Address

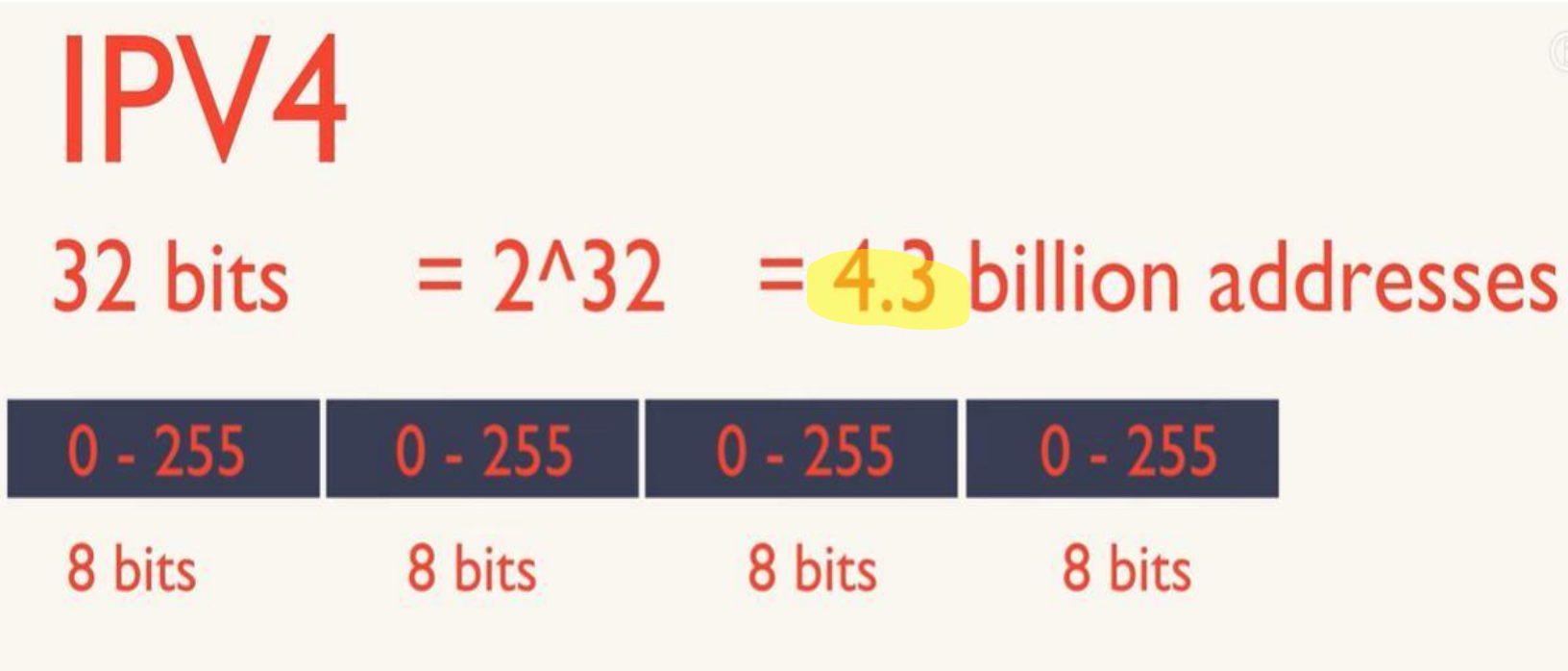
Structure of an IPv4 Address

- Total bits: **32 bits**
- Divided into **4 octets** (8 bits each)
- Each octet ranges from **0–255**
- **Example:**

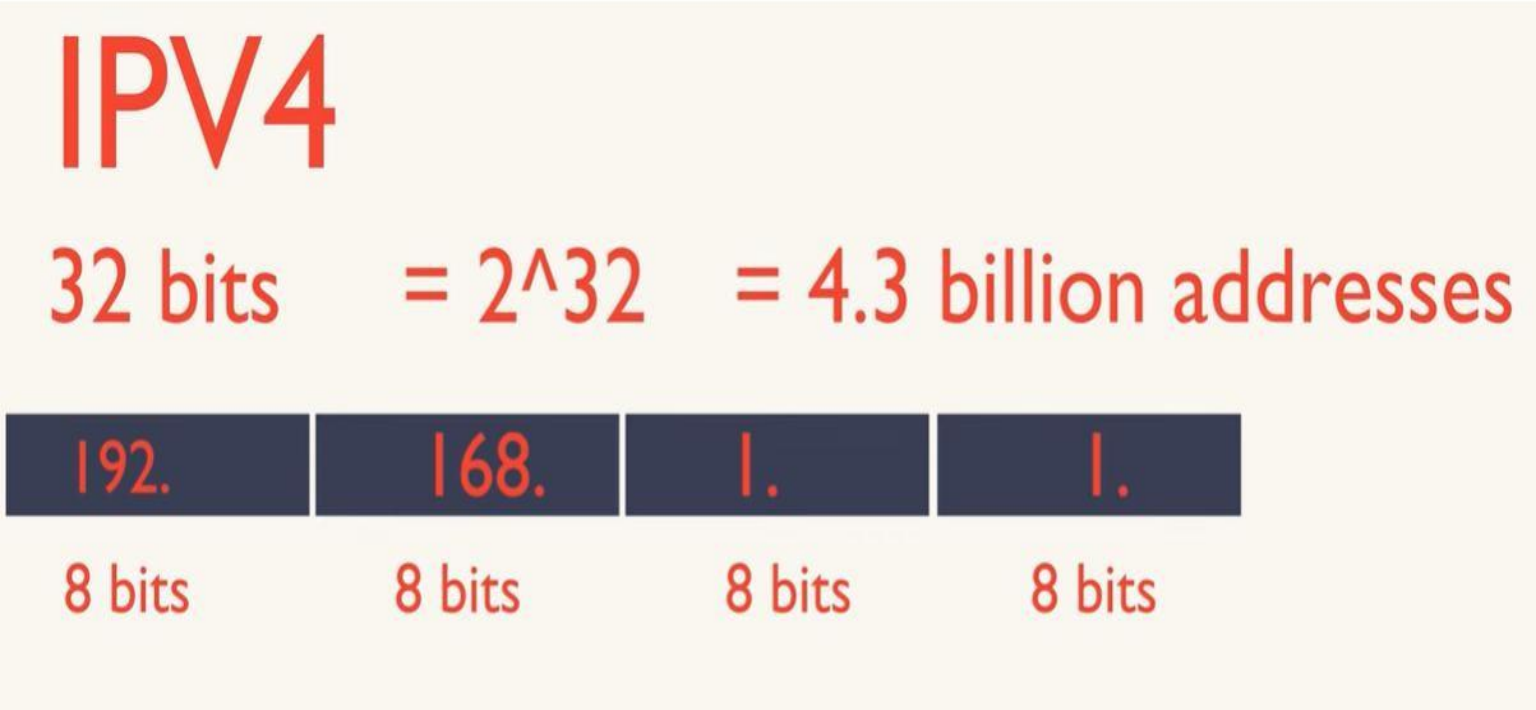
11000000.10101000.00000001.00000001

192 . 168 . 1 . 1

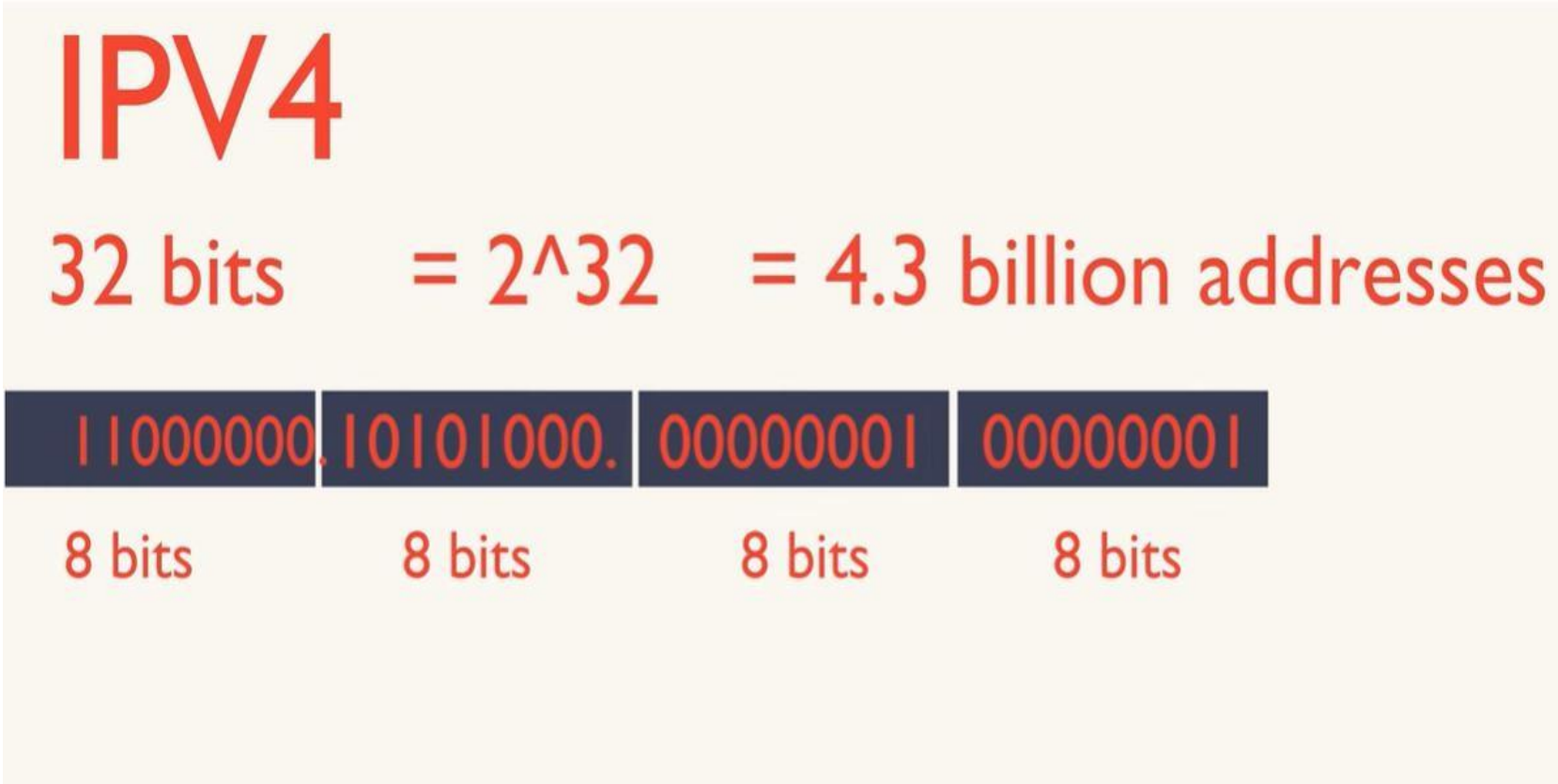
IPV4 address



IPV4 address



IPV4 address



IPV4 address

IPV4 address

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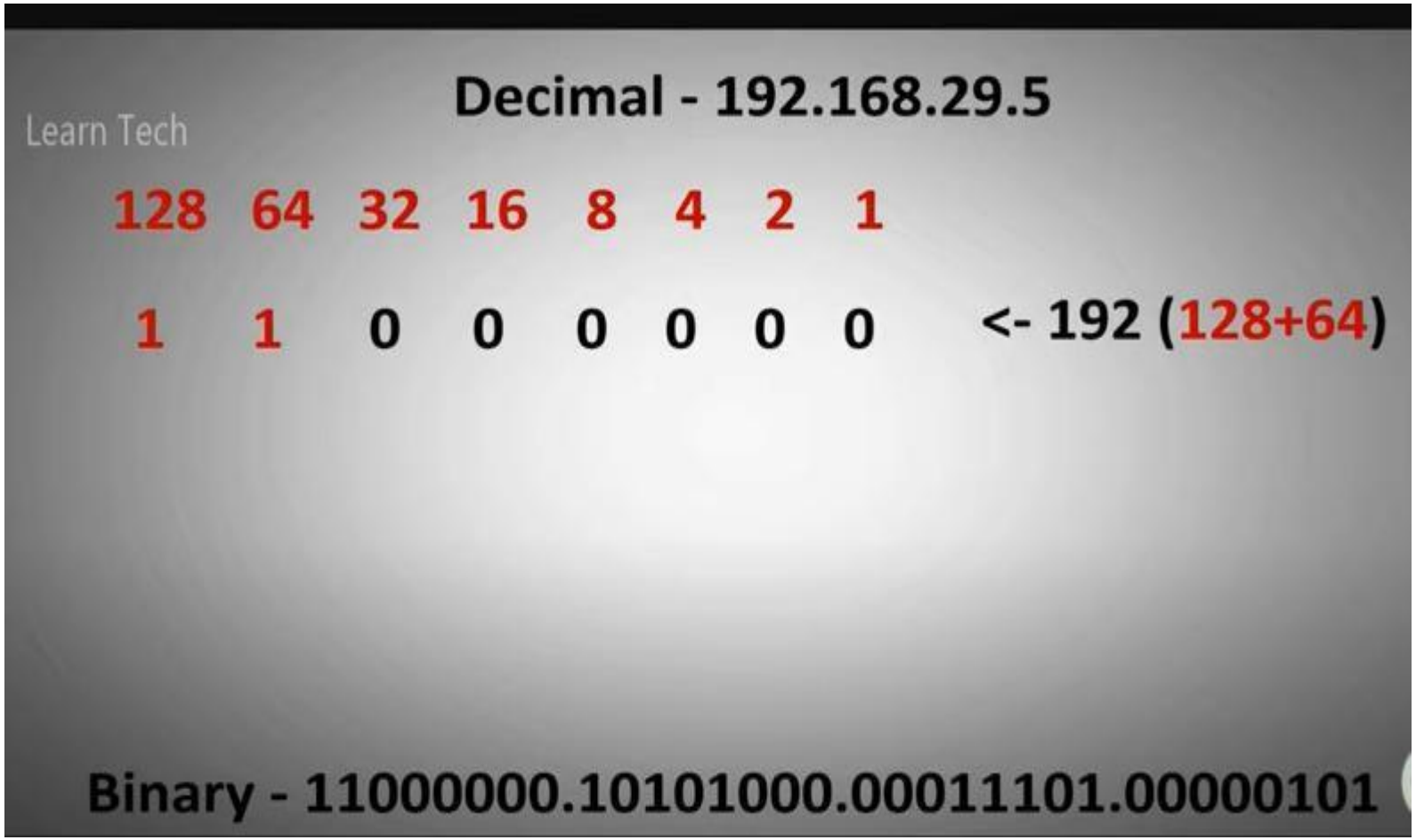

IPV4 address

Octet	Powers of 2	Calculation	Binary
192	128,64,32,16,8,4,2,1	128 + 64 = 192	11000000
168	128,64,32,16,8,4,2,1	128 + 32 + 8 =168	10101000
29	128,64,32,16,8,4,2,1	16 + 8 + 4 + 1=29	00011101
5	128,64,32,16,8,4,2,1	4 + 1 = 5	00000101

11000000.10101000.00011101.00000101

Classes of IPv4 Addresses

S.NO	CLASS	RANGE	DEFAULT SUBNET MASK	USAGE
1.	A	1.0.0.0 – 126.0.0.0	255.0.0.0	Large networks
2.	B	128.0.0.0 – 191.255.0.0	255.255.0.0	Medium networks
3.	C	192.0.0.0 – 223.255.255.0	255.255.255.0	Small networks
4.	D	224.0.0.0 – 239.255.255.255	–	Multicast
5.	E	240.0.0.0 – 255.255.255.255	–	Experimental

Types of IPv4 Addresses

- **Public IP – Used on the internet**
- **Private IP – Used inside local networks**
  - 10.0.0.0 – 10.255.255.255
  - 172.16.0.0 – 172.31.255.255
  - 192.168.0.0 – 192.168.255.255
- **Static IP – Manually assigned**
- **Dynamic IP – Assigned automatically (DHCP)**

Limitations of IPv4

- Limited address space (≈ 4.3 billion)
- Address exhaustion
- Not suitable for massive IoT growth
- This led to the development of IPv6

Summary

- IPv4 is used to uniquely identify devices on a network
- It uses a 32-bit addressing format written in dotted decimal form
- IPv4 includes public, private, static, and dynamic IP addresses
- NAT helps multiple devices access the internet using one public IP
- IPv4 is the backbone of today's internet communication, but it is reaching its limits